Thesis Progress - Writing

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Wave Functions in Quantum Cascade Lasers Using the Transfer Matrix Method

1 Introduction

Quantum mechanics is the branch of physics responsible for studying the behavior of matter and energy at extremely small scales, such as atoms, electrons, and other subatomic particles.

Its development made it possible to understand nature in a way that differed considerably from the understanding provided by classical mechanics. Unlike classical mechanics, in which the motion of a particle can be described through quantities such as its position and velocity, quantum mechanics establishes that the behavior of particles must be described by means of a quantum state. This state can be represented by a wave function, whose interpretation makes it possible to determine the probability of finding a particle in a given region of space.

The development of quantum mechanics made it possible to explain various phenomena that could not be adequately described by classical theories. Some examples are blackbody radiation, whose study led to the proposal of energy quantization; the photoelectric effect, through which the interaction between light and matter could be understood; and the electronic structure of atoms, which made it possible to explain the existence of discrete energy levels. These phenomena were important in establishing the foundations of quantum physics and understanding the behavior of matter at microscopic scales.

These advances fundamentally changed the understanding of matter and gave rise to the development of new research areas and technologies based on quantum phenomena. Their applications include transistors, light-emitting diodes, lasers, electron microscopes, and various devices used in diagnostic and imaging systems. Thus, the principles of quantum mechanics not only make it possible to explain phenomena that occur at microscopic scales, but are also used in the development of technologies that are part of everyday life.

One of the areas in which the principles of quantum mechanics are particularly important is the study of semiconductor materials. The electronic properties of these materials can be modified by combining different compositions and by fabricating structures with dimensions small enough to produce quantum confinement effects. Under these conditions, the motion of electrons no longer behaves as a continuous system, and discrete energy states appear that depend on the characteristics of the structure in which they are confined.

Quantum confinement can be produced by constructing semiconductor heterostructures, which are formed by layers of different materials with distinct electronic properties. When a layer of a semiconductor material with a lower conduction-band energy is surrounded by materials with a higher energy in that band, a potential region capable of confining electrons is generated. These structures are known as quantum wells and are among the fundamental elements in the development of various optoelectronic devices.

The study of electrons confined in these systems requires determining the allowed energy states and their corresponding wave functions. These functions contain information about the spatial distribution of electronic states and make it possible to identify the regions in which there is a greater probability of finding electrons. For this reason, determining the wave functions is fundamental to understanding and designing devices based on nanoscale semiconductor structures.

Quantum cascade lasers (QCLs) belong to this context. They are unipolar semiconductor devices whose operation is based on quantum confinement phenomena and electronic transitions between quantized states in the conduction band. Their development represented an important application of semiconductor heterostructure engineering because it makes it possible to design the emission properties of the device by modifying its structural characteristics.

The first quantum cascade laser was developed in 1994 by J. Faist and collaborators at Bell Laboratories. Since then, these devices have become important sources of radiation in the mid- and far-infrared regions, as well as in the terahertz region. Their characteristics have enabled the exploration of applications related to spectroscopy, gas detection, environmental monitoring, medical diagnosis, imaging systems, and communications, among others.

Unlike conventional semiconductor lasers, whose operation is based on interband transitions between the conduction band and the valence band, QCLs use intersubband transitions between quantized states within the conduction band. This characteristic means that the energy of the electronic transition does not depend exclusively on the material's band-gap energy, but can instead be controlled through the design of the heterostructure, particularly through the thicknesses and compositions of the quantum wells and barriers.

The active region of a QCL consists of a sequence of quantum wells and barriers with nanometer-scale dimensions. The layers that make up this region are designed so that electrons can occupy specific energy states and undergo transitions between them. In addition, the barriers separating the wells can allow electron transport through the tunneling effect, making it possible for an electron to pass successively through different regions of the structure.

This arrangement gives rise to what is known as the cascade effect. After undergoing a radiative transition between two electronic states and emitting a photon, the electron can relax to a lower-energy state and subsequently be transferred through tunneling to another period of the structure. The process can be repeated across multiple periods, allowing the same electron to participate in the generation of several photons. In this way, the design of the active region significantly determines the electronic and optical properties of the device.

Because the properties of electronic states depend directly on the characteristics of the layers that make up the heterostructure, methods are needed to adequately study the behavior of electrons within these devices. Parameters such as layer thickness, material properties, and the applied electric potential can modify the energy levels and the distribution of the wave functions. Even small variations in layer thickness can produce significant changes in the electronic states of the structure.

To determine these states, it is necessary to solve the Schrödinger equation while considering the characteristics of the heterostructure and the conditions to which the device is subjected. However, because QCLs consist of multiple layers with different properties, solving the problem can become complex and may require numerical methods that provide adequate accuracy at a reasonable computational cost.

Among the methods used to solve problems of this type is the transfer matrix method, which makes it possible to study one-dimensional structures formed by different potential regions. This method consists of relating the wave function across the different layers that make up the structure, making it possible to obtain the energy values corresponding to the allowed states and their respective wave functions.

In this work, the transfer matrix method is proposed for studying the wave functions of a quantum cascade laser structure. The BenDaniel-Duke model, based on the use of effective masses and focused on the study of the conduction band, is considered to describe the electronic behavior. The structure will be represented by discretized regions in which the potential can be considered constant, allowing the corresponding transfer matrices to be constructed for each region.

The search for energy states will be carried out using the shooting method, complemented by the bisection method to find the energy values that satisfy the conditions required for the quantum states of the structure. From these values, it will be possible to obtain the wave functions and analyze their behavior within the wells and barriers that make up the heterostructure.

Finally, the results obtained through the numerical procedure will be evaluated by comparison with analytical solutions corresponding to simple potentials. The stability of the solution will also be analyzed as the layer thicknesses are modified and the number of regions making up the structure increases. The aim is to determine the accuracy of the transfer matrix method for studying quantum cascade laser structures and to establish a foundation for developing a simulator that can analyze different configurations of these devices.

2 Problem Context

Quantum cascade lasers consist of semiconductor heterostructures made up of multiple layers of quantum wells and barriers. The operation of these devices depends on the configuration of these layers, since parameters such as their thicknesses, the materials used, and the applied electric potential directly influence the energy levels and the wave functions of the electrons.

Because the layers have nanometer-scale dimensions, small variations in their thicknesses can produce significant changes in the electronic states of the structure. For this reason, before fabricating a QCL, it is necessary to have tools that make it possible to study its behavior and determine whether the proposed configuration has the desired characteristics.

To obtain the energy levels and wave functions, it is necessary to solve the Schrödinger equation while considering the different regions that make up the heterostructure. Although analytical solutions may exist for simple systems, structures with multiple wells and barriers require the use of numerical methods. Among these methods is the transfer matrix method, which makes it possible to relate the wave function across the different layers of the structure.

However, as the number of layers increases, the solution may present difficulties related to numerical accuracy and stability. In addition, the energy values corresponding to bound states must be determined through search procedures. Therefore, it is necessary to evaluate the behavior of the method used under different configurations of the structure.

In this context, the development of a computational procedure is proposed to obtain the energy levels and wave functions of a quantum cascade laser structure using the transfer matrix method, considering the BenDaniel-Duke model and employing the shooting and bisection methods to search for the energy states.

3 Problem Definition

Studying a quantum cascade laser structure requires performing a series of calculations to determine the energy levels and wave functions of the electrons present in its different quantum wells and barriers. These calculations can become complex when the number of layers, their thicknesses, the materials used, and the applied potential are modified, since each change can generate new electronic states that must be determined.

Performing this process manually for each configuration of the structure involves repeating a large number of calculations, which can make the analysis more difficult and increase the time required to study different configurations. In addition, analyzing the behavior of the wave functions requires the use of complex numerical procedures that provide results with adequate accuracy.

Given this situation, there is a need to implement these procedures computationally so that the process of obtaining results can be automated and accelerated. A computational tool will make it possible to modify the structure's parameters and obtain the corresponding results without having to repeat the entire procedure manually.

Therefore, the problem addressed by this research consists of developing a computational procedure based on the transfer matrix method that makes it possible to optimize the calculation of the energy levels and wave functions of a quantum cascade laser structure by solving the Schrödinger equation while considering the BenDaniel-Duke effective-mass model. Likewise, the aim is to evaluate the accuracy and stability of the results obtained through the developed procedure.

4 General Objective

To develop a computational procedure in the C++ programming language that makes it possible to create a simulation of a Quantum Cascade Laser structure and determine its energy levels and wave functions by solving the Schrödinger equation using the Transfer Matrix Method and considering the BenDaniel-Duke effective-mass model, with the ultimate purpose of analyzing the accuracy and stability of the results obtained.

5 Specific Objectives